Build an Environmental Station

Sensor light → hello screen → live station → fault and recovery

Keep each win. Add only the wires that unlock the next one.

Lesson	015 — project	**Time**	about 75 minutes
Prerequisites	Lessons 001–014	**Board**	Arduino Mega 2560
Visible story	D13, LCD, RGB	**Power**	USB only
Finish	two-minute rest, then RESET replays everything		

Build safely. Unplug USB before every wiring change. This project conditionally uses two independently identified parts: a labeled 5 V DHT11 module and a parallel 16-pin LCD1602 whose pins are documented as VSS, VDD, VO, RS, RW, E, D0–D7, A, K. A retail family listing does not establish either pin order. Leave LCD A and K open unless the exact backlight rating and resistor are known. Use a common-cathode RGB LED with one 330 Ω resistor per die. Stop for heat, odor, resets, or a wire that disagrees with a printed label.

1 The four-win plan

Start from the climate light in lesson 013 and the LCD wiring knowledge from lesson 014. The final sketch does the quiet engineering work, but the build feels like a sequence of visible rewards:

1. the existing DHT11 and RGB light produce a valid climate color;
2. D13 flashes, then the correctly wired LCD says `STATION STARTING`;
3. the first complete sensor frame becomes a numbered LCD record and the RGB color reacts to humidity;
4. lifting DATA while unpowered creates a red fault rhythm; restoring it lets the next good frame recover automatically.

After two minutes the screen says `STATION RESTING`, then every owned pin is released. Press RESET to replay the whole story. There is no worksheet or terminal ritual between those wins.

2 Plate 1 — Find every Mega socket

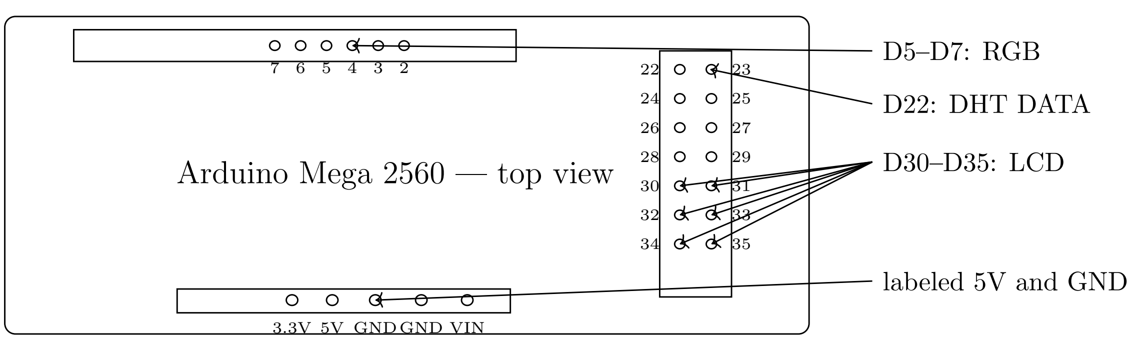

Figure 1: Mega locator plate. D22 and D30–D35 are sockets in the paired vertical digital header; D5–D7 are in the top digital header. D13 is the built-in acquisition LED.

Touch each printed socket label with USB unplugged. In particular, do not mistake D22 for the top-header socket marked 2.

3 Plate 2 — Keep the working climate light

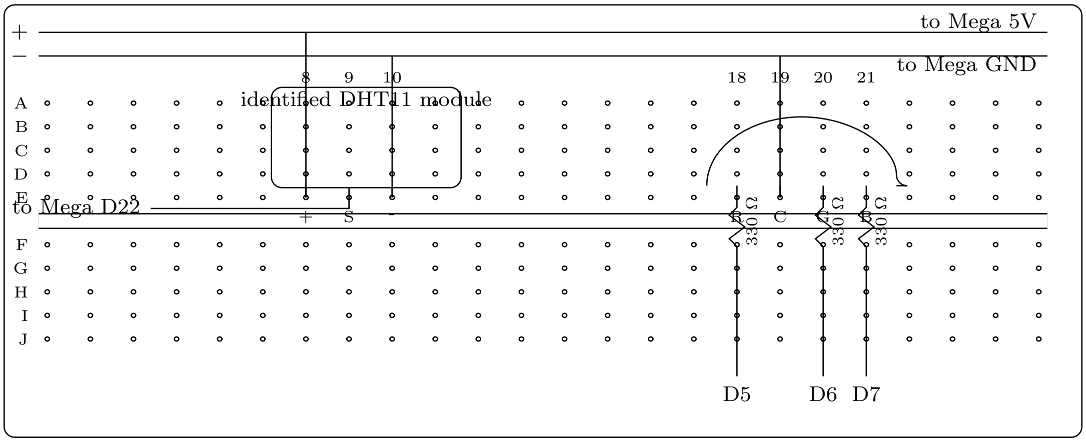

Figure 2: Literal first breadboard. Columns run left to right; A–E and F–J are separate five-hole groups. The module's *named* +, S/DATA, and − leads land at E8–E10 even if its physical lead order requires short re-routing jumpers. The identified common-cathode RGB legs land at E18–E21.

From	To	Purpose
Mega 5V / GND	breadboard + / − rails	one supply and return
module +, DATA, −	E8 to +, E9 to D22, E10 to −	route by printed names, not guessed order
RGB red, green, blue	E18/E20/E21 through 330 Ω to D5/D6/D7	three bounded color channels
RGB common cathode	E19 to − rail	common return

Replay lesson 013's climate-light sketch now. The RGB light should settle into a climate color after a valid frame. Keep USB unplugged after that win while you add the LCD; upload the lesson 015 project only after Plate 3 is complete.

You unlocked: the DHT11 can produce a complete frame and the RGB LED can show it.

4 Plate 3 — Add the LCD on the right half

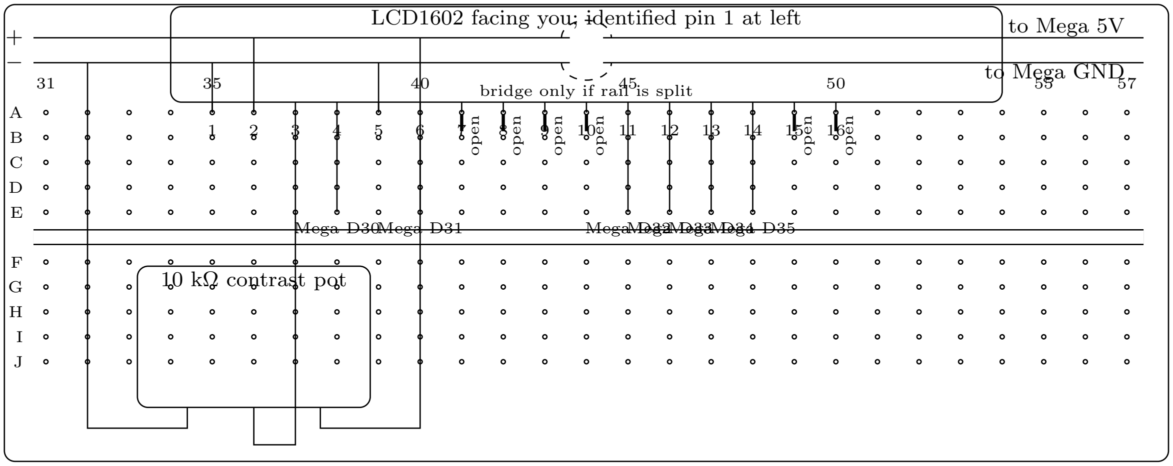

Figure 3: Literal close-up of columns 31–57 on the *same* 830-point breadboard. LCD pins 1, 2, 3, 4, 5, 6, and 11–14 connect; pins 7–10 remain open in four-bit mode; pins 15–16 remain open because this lesson makes no backlight assumption. The contrast potentiometer ends reach the rails and its center wiper reaches LCD pin 3. Dashed jumpers bridge a rail split only when an unpowered continuity check finds one.

LCD pin	Name	Literal connection
1, 2, 3	VSS, VDD, VO	GND, 5 V, contrast-pot wiper
4, 5, 6	RS, RW, E	Mega D30, GND, Mega D31
7–10	D0–D3	leave open
11–14	D4–D7	Mega D32, D33, D34, D35 in order
15–16	A, K	leave open unless independently qualified later
rail split	+ and −	same-polarity bridges only if continuity shows a break

Keep the climate light in the left-side columns from Plate 2 and place this LCD close-up in right-side columns 31–57 of the same 830-point board. With USB unplugged, check continuity along both power rails. If either rail is split, fit the two dashed same-polarity bridges shown; never bridge + to −. Set contrast near its midpoint. Inspect every numbered path, plug in USB, and press RESET.

You unlocked: D13 flashes, then the LCD says `STATION STARTING` and `WAIT FOR RECORD`. That screen requires all sensor, LCD, RGB, and D13 resources to have been acquired.

5 Plate 4 — Watch the station tell its story

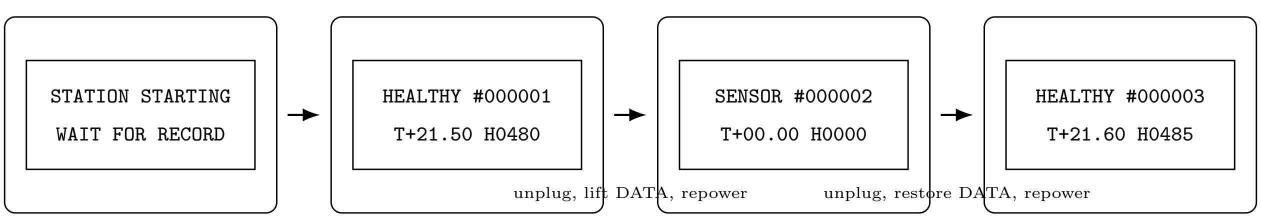

Figure 4: Visible state progression. Exact values vary, but success requires a numbered healthy record and a climate color. Missing DATA produces an explicit fault instead of counterfeit values; restoring DATA permits a later healthy record.

The RGB LED is part of the story:

Light	Meaning
blue pulse	waiting for a first complete frame
steady green	valid humidity below 55% RH
steady cyan	valid humidity from 55% through 69.9% RH
steady amber	valid humidity at least 70% RH
two red pulses	sensor transaction or validation fault
three red pulses	impossible caller time; RESET starts a new bounded run

Cup your hands loosely near the sensor or breathe gently across it from several inches away. Do not wet or touch the vents. Watch the displayed humidity and color move together. A complete frame, checksum, range check, and freshness rule stand between a wire twitch and a healthy record.

For the reversible surprise, unplug USB, lift only the D22-to-DATA jumper, and repower. The station keeps running but shows a named sensor fault with red pulses. Unplug, restore that jumper, and repower. The hello, first record, and climate color return. This proves recovery by producing the cool result, not by filling out a test card.

6 What the code protects

```
const adk::Status                status   = station.update (now);
const adk::EnvironmentalSnapshot evidence = station.snapshot ();

presentEnvironment (evidence);
showHealth (evidence.record.health, now);
```

`EnvironmentalStation` owns the sensor lifecycle and samples only when its caller time says a record is due. A late loop makes one observation, not a burst of stale transactions. A record keeps its values, sample state, sensor status, timestamp, and sequence together. Invalid or missing sensor data stays explicit; this is a learning instrument, not a calibrated monitor or alarm.

The LCD and RGB LED are separate presentation objects. A blank LCD cannot rewrite sensor truth, and an old good sample cannot masquerade as a new one. After 120 seconds, the station displays its rest message for 1.5 seconds, turns off the indicators, shuts down the LCD and sensor, and releases D13, D5–D7, D22, and D30–D35. RESET begins another run.

Behind the scenes, automated host checks still replay exact deadlines, transport faults, stale data, recovery, resource conflicts, rollback, timestamp rollover, restart, repeated shutdown, and identical traces. Those checks keep the code sturdy; they are not learner chores.

7 If a win does not appear

What you see	Likely boundary	Next move with USB unplugged
no D13 flash	acquisition or power	inspect all named pins for duplicates or shorts
D13 flash, blank LCD	LCD supply, contrast, or pin order	verify pins 1–6 and 11–14 by number; turn contrast gently
garbled text	D4–D7 order	trace LCD 11, 12, 13, 14 to D32, D33, D34, D35
hello forever	DHT path	verify named module leads, D22, supply, and documented pull-up
record but RGB dark	LED identity or channel path	verify common cathode and all three 330 Ω resistors
red pulses	missing or rejected sensor frame	restore DATA and shorten/check the identified module wiring
plausible but surprising values	sensor limitations or environment	treat them as educational indications, not calibrated facts

8 Make it yours

1. Change the two starting lines to a family station name, keeping each line at 16 printable characters or fewer.
2. Choose three different low-duty RGB colors for dry, comfortable, and humid air, then predict which one appears before RESET.
3. Add a tiny arrow or face to the second LCD row without hiding the record number or the explicit fault word.
4. Press RESET near the two-minute boundary and watch the complete acquisition-to-record story begin again.

9 Next

Keep the delight: one correctly connected subsystem unlocks the next visible state. Lesson 016 changes the input side of that pattern by turning a keypad matrix into explicit operator events.